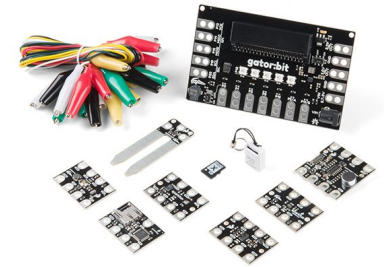


Micro:bit Gator Science Kit

WHAT'S IN THE KIT

- Gator boards with audio output
- UV sensor
- Soil moisture sensor
- Environmental sensors
- NOTE: requires accessories from a Micro:bit Kit — best checked out together



QUICK-START GUIDE

1. Check out a Micro:bit Kit alongside this one.
2. Clip a gator board to the micro:bit with alligator clips.
3. In MakeCode, add the gator:bit extension for sensor blocks.
4. Code readings to the LED display or log them over time.
5. Head outside and start measuring!

FULL LESSONS IN THIS GUIDE

GRADES 3-5	Thirsty Plants Investigation	page 2
GRADES 6-8	Playground Microclimates	page 3
GRADES 9-12	Environmental Data Logger	page 4

PRO TIP: Best paired with a Micro:bit Kit checkout — the gator boards need the micro:bits and clips.

Micro:bit Gator Science Kit

LESSON · GRADES 3-5

Thirsty Plants Investigation

LEARNING OUTCOMES

- Students will use a digital sensor to collect environmental data
- Students will compare measurements across conditions and record findings
- Students will draw evidence-based conclusions about plant needs

MATERIALS

- Gator Science boards + micro:bits with clips (from Micro:bit Kit)
- Classroom plants or soil samples in cups
- Data recording sheet
- Chromebooks with MakeCode + gator:bit extension

INSTRUCTIONS

1. Pre-load the soil moisture program or build it together with the class.
2. Model how to probe soil and read the value on the LED display.
3. Teams measure every classroom plant and record values.
4. Rank plants from driest to wettest; predict which needs water first.
5. Water one plant, re-measure, and quantify the change.
6. Assign a data-based watering schedule as the takeaway product.

ACTIVITY

Thirsty Plants Investigation — students probe classroom plants, rank them by moisture reading, and publish a watering schedule backed by their own sensor data. Re-measuring after watering closes the loop.

Micro:bit Gator Science Kit

LESSON · GRADES 6-8

Playground Microclimates

LEARNING OUTCOMES

- Students will plan a controlled outdoor data collection with multiple sensors
- Students will graph environmental data and identify patterns
- Students will construct an evidence-based argument from collected data

MATERIALS

- Gator boards + micro:bits (1 set per team)
- UV and temperature sensing set up in MakeCode
- Clipboards and site maps of the school grounds
- Sheets/graph paper for analysis

INSTRUCTIONS

1. Teams mark 4-5 measurement sites on a playground map (sun, shade, blacktop, grass).
2. Build or load the UV + temperature logging program.
3. Field session: measure each site twice at consistent heights; record on clipboards.
4. Back inside, graph readings by site and look for microclimate patterns.
5. Each team drafts a claim ('the new bench should go here because...') supported by data.
6. Teams present competing proposals; class votes on the strongest evidence.

ACTIVITY

Playground Microclimates — teams map UV and temperature across the school grounds, then argue from evidence where a bench, garden, or shade sail belongs. Real site, real stakes, real data.

Micro:bit Gator Science Kit

LESSON · GRADES 9-12

Environmental Data Logger

LEARNING OUTCOMES

- Students will design a multi-day environmental logging experiment
- Students will export, clean, and analyze time-series sensor data
- Students will communicate findings with appropriate graphs and uncertainty

MATERIALS

- Gator boards + micro:bits per team
- MakeCode data-logging setup
- Google Sheets for analysis
- Lab notebook or digital doc for the experimental plan

INSTRUCTIONS

1. Teams write an experimental plan: question, variables, sites, sampling schedule.
2. Review plans for controls and consistency before any data collection.
3. Run the logging protocol over 2-3 days (soil, UV, temperature at set times).
4. Import readings into Sheets; clean anomalies and note their causes.
5. Produce trend graphs and a short written analysis with limitations.
6. Present as a mini poster session; peers critique methods, not just results.

ACTIVITY

Environmental Data Logger — a full data-science cycle: students design the protocol, run multi-day collection with hardware they configured, and defend their analysis including what their data can't say.